Phoenix HeatShield - Idea Proposal

Samsung Innovation Campus - Group 13 - 12 November 2026

1. Project Idea

Extreme heat affects neighborhoods differently. Tree cover, built surfaces, income, and vehicle access can influence how much heat people face and how well they can cope. We plan to build a web application for 378 census tracts in Phoenix, Arizona. It will combine satellite, environmental, and socioeconomic data to model daytime surface temperature and show which areas should be considered first for cooling efforts.

2. Relevance to Sustainable Development Goals (SDGs)

SDG 11 - Sustainable Cities and Communities. Help cities plan for extreme heat.
SDG 13 - Climate Action. Support local adaptation to rising temperatures.
SDG 10 - Reduced Inequalities. Identify areas where limited resources may make it harder to cope with heat.

3. Literature Examples

Bautista et al. (2022) used satellite and community data to identify suitable locations for tree planting in Phoenix. This gives us an example of combining heat and social indicators to set priorities [1].
Li et al. (2016) studied how land cover and socioeconomic conditions relate to surface temperature in Phoenix. Their findings help us choose variables for our model [2].

4. Describe Your Data

We have data for 378 census tracts in GeoTIFF, CSV, and GeoJSON formats. NASA MODIS MOD11A2.061 provides 1 km daytime land surface temperature and quality flags for nine eight-day composites starting between May 25 and July 28, 2023. CAP LTER provides 30 m NDVI and ENDISI rasters. Phoenix GIS provides tree cover, income, and vehicle-access data. Preprocessing will include coordinate alignment, seasonal-band selection, quality filtering, tract-level aggregation, missing-value handling, and checking spatial joins.

5. Approach (Machine Learning or Deep Learning)

We will use **Random Forest Regression**, a machine learning method suited to our small tabular dataset and nonlinear relationships. The model will predict daytime surface temperature from environmental variables. We will compare it with linear regression using spatial cross-validation, MAE, and RMSE. SHAP will help explain the predictions. We will then combine heat exposure with income and vehicle access to rank areas, and check how changing indicator weights affects this ranking.

References

[1] Bautista, A., et al. (2022). *Employing NASA Earth Observations to Conduct Site Suitability Analyses on Residential Tree Planting Initiatives in Phoenix, AZ.* NASA DEVELOP technical report.

[2] Li, X., et al. (2016). *Remote sensing of the surface urban heat island and land architecture in Phoenix, Arizona.* Remote Sensing of Environment, 174, 233-243. doi:10.1016/j.rse.2015.12.022.